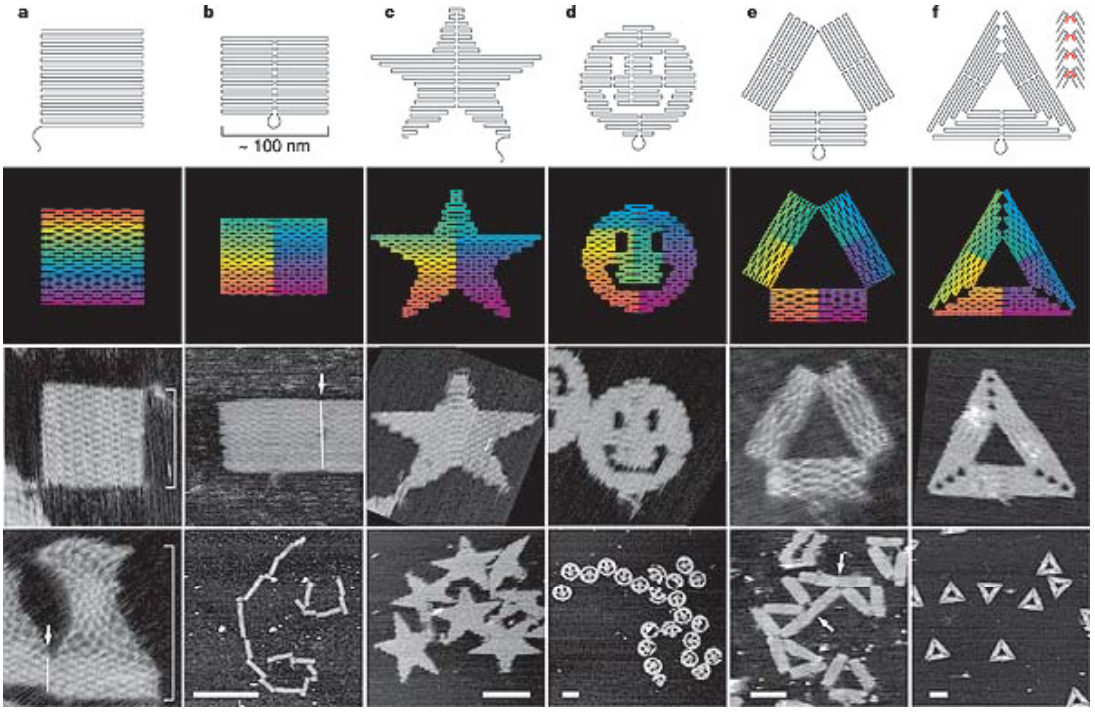
# 화학특강 (2011년 2학기)
# 제2장 DNA, 단백질 그리고 Biotechnology

중요 학습 내용

- DNA의 구조
- 단백질의 구조와 합성
- DNA translocation and DNA transfection
- Colorimetric sensing



## DNA Origami

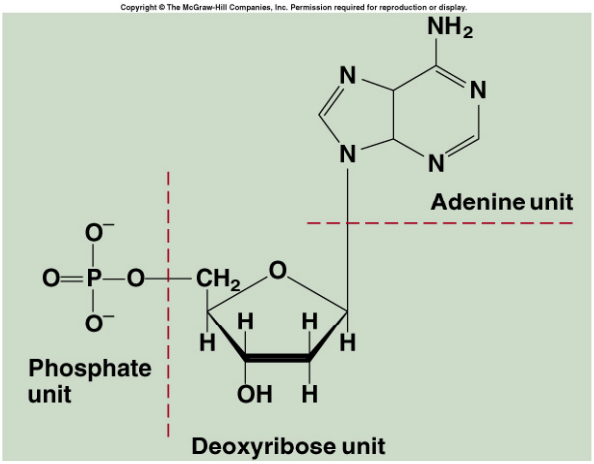



Rothemund, Nature **440**, 297 (2006)

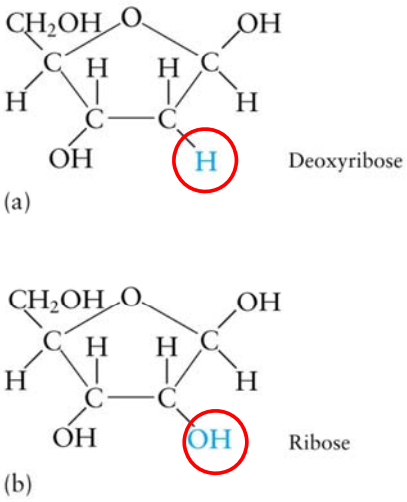
# Structures of Deoxyribonucleic Acid (DNA) and Ribonucleic Acid (RNA)

- Structure of Nucleotide

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- Deoxyribose and ribose

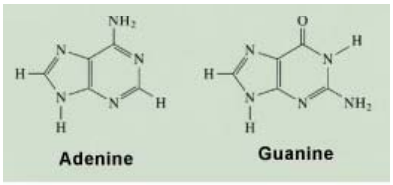



Chang, 25.4

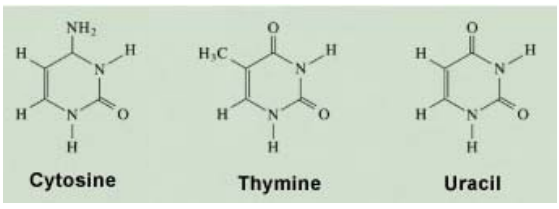


# Structures of Deoxyribonucleic Acid (DNA) and Ribonucleic Acid (RNA)

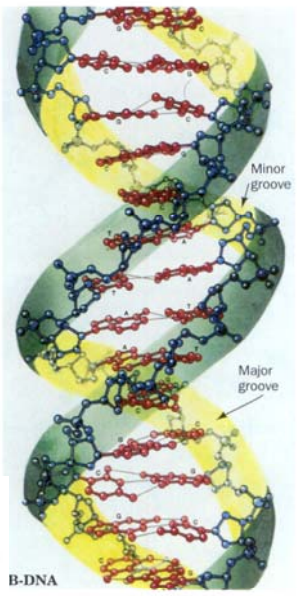
- Purine bases

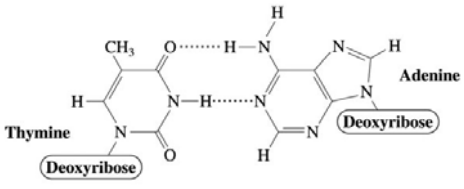


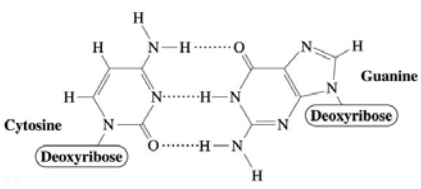

- Pyrimidine bases




- DNA double helix



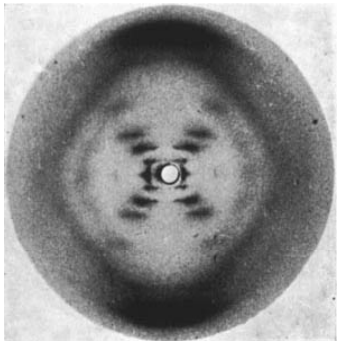




Chang, 25.4

# History of DNA

- Chargaff's rules in the 1940's
  1. The amount of adenine (a purine) is equal to that of thymine (a pyrimidine); that is A = T, or A/T = 1.
  2. The amount of guanine (a purine) is equal to that of cytosine (a pyrimidine); that is G = C, or G/C = 1.
  3. The total number of purine bases is equal to the total number of pyrimidine bases; that is, A + G = T + C .

- X-ray diffraction photograph by R. Franklin





Chang, 25.4



# History of DNA

- Nature paper by Watson & Crick

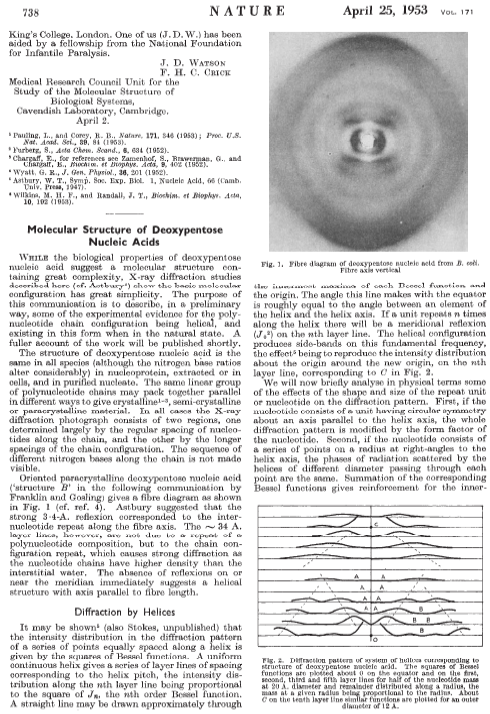


MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

Watson & Crick, Nature 171, 737 (1953)

- Nature paper by Wilkins



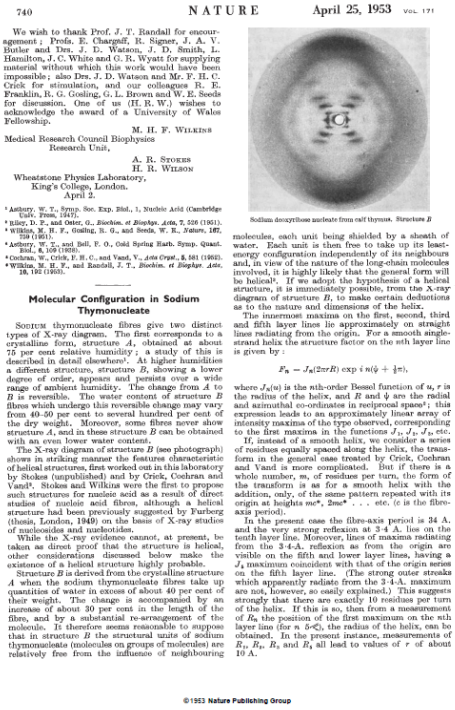
Molecular Structure of Deoxypentose Nucleic Acids

Wilkins, Stoke, and Wilson, Nature 171, 738 (1953)

©1953 Nature Publishing Group

# History of DNA

- Nature paper by Rosalind Franklin



740 NATURE April 25, 1953 Vol. 171

We wish to thank Prof. J. T. Randall for encouragement; Profs. E. Chargaff, R. Signer, J. A. V. Butler and Drs. J. D. Watson, J. D. Smith, L. Hamilton, J. C. White and G. R. Wyatt for supplying material without which this work would have been impossible; also Drs. J. D. Watson and Mr. F. H. C. Crick for stimulation, and our colleagues R. E. Franklin, R. G. Gosling, G. L. Brown and W. E. Seeds for discussion. One of us (H. R. W.) wishes to acknowledge the award of a University of Wales Fellowship.

M. H. F. Wilkins
Medical Research Council Biophysics Research Unit,

A. R. Stokes
H. R. Wilson
Wheatstone Physics Laboratory, King's College, London.
April 2.

Sodium deoxyribose nucleate from calf thymus. Structure B

**Molecular Configuration in Sodium Thymonucleate**

Sodium thymonucleate fibres give two distinct types of X-ray diagram. The first corresponds to a crystalline form, structure A, obtained at about 75 per cent relative humidity; a study of this is described in detail elsewhere[3]. At higher humidities a different structure, structure B, showing a lower degree of order, appears and persists over a wide range of ambient humidity. The change from A to B is reversible. The water content of structure B fibres which undergo this reversible change may vary from 40–50 per cent to several hundred per cent of the dry weight. Moreover, some fibres never show structure A, and in these structure B can be obtained with an even lower water content.

The X-ray diagram of structure B (see photograph) shows in striking manner the features characteristic of helical structures, first worked out in this laboratory by Stokes (unpublished) and by Crick, Cochran and Vand[4]. Stokes and Wilkins were the first to propose such structures for nucleic acid as a result of direct studies of nucleic acid fibres, although a helical structure had been previously suggested by Furberg (thesis, London, 1949) on the basis of X-ray studies of nucleosides and nucleotides.

While the X-ray evidence cannot, at present, be taken as direct proof that the structure is helical, other considerations discussed below make the existence of a helical structure highly probable.

Structure B is derived from the crystalline structure A when the sodium thymonucleate fibres take up quantities of water in excess of about 40 per cent of their weight. The change is accompanied by an increase of about 30 per cent in the length of the fibre, and by a substantial re-arrangement of the molecule. It therefore seems reasonable to suppose that in structure B the structural units of sodium thymonucleate (molecules or groups of molecules) are relatively free from the influence of neighbouring molecules, each unit being shielded by a sheath of water. Each unit is then free to take up its least-energy configuration independently of its neighbours and, in view of the nature of the long-chain molecules involved, it is highly likely that the general form will be helical[5]. If we adopt the hypothesis of a helical structure, it is immediately possible, from the X-ray diagram of structure B, to make certain deductions as to the nature and dimensions of the helix.

The innermost maxima on the first, second, third and fifth layer lines lie approximately on straight lines radiating from the origin. For a smooth single-strand helix the structure factor on the nth layer line is given by:

$$F_n = J_n(2\pi rR)\exp i n(\psi + \tfrac{1}{2}\pi),$$

where $J_n(u)$ is the nth-order Bessel function of $u$, $r$ is the radius of the helix, and $R$ and $\psi$ are the radial and azimuthal co-ordinates in reciprocal space[6]; this expression leads to an approximately linear array of intensity maxima of the type observed, corresponding to the first maxima in the functions $J_1$, $J_2$, $J_3$, etc.

If, instead of a smooth helix, we consider a series of residues equally spaced along the helix, the transform in the general case treated by Crick, Cochran and Vand is more complicated. But if there is a whole number, $n$, of residues per turn, the form of the transform is as for a smooth helix with the addition, only, of the same pattern repeated with its origin at heights $n\sigma$, $2n\sigma$, ... etc. ($\sigma$ is the fibre-axis period).

In the present case the fibre-axis period is 34 A. and the very strong reflexion at 3·4 A. lies on the tenth layer line. Moreover, lines of maxima radiating from the 3·4-A. reflexion as from the origin are visible on the fifth and lower layer lines, having a $J_4$ maximum coincident with that of the origin series on the fifth layer line. (The strong outer streaks which apparently radiate from the 3·4-A. maximum are not, however, so easily explained.) This suggests strongly that there are exactly 10 residues per turn of the helix. If this is so, then from a measurement of $R_4$ the position of the first maximum on the nth layer line (for $n$ 5-6), the radius of the helix, can be obtained. In the present instance, measurements of $R_4$, $R_5$, $R_6$ and $R_4$ all lead to values of $r$ of about 10 A.

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**Franklin & Gobling, Nature 171, 740 (1953)**

## The Nobel Prize in Physiology or Medicine 1962

"for their discoveries concerning the molecular structure of nucleic acids and its significance for information transfer in living material"



**Francis Harry Compton Crick**

1/3 of the prize

United Kingdom

MRC Laboratory of Molecular Biology
Cambridge, United Kingdom



**James Dewey Watson**

1/3 of the prize

USA

Harvard University
Cambridge, MA, USA



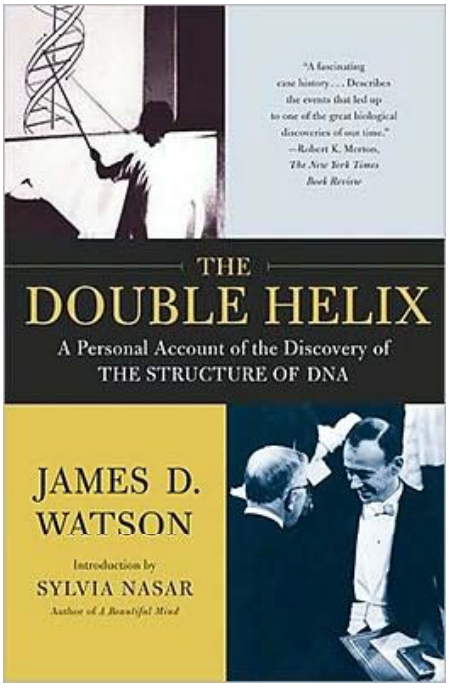
**Maurice Hugh Frederick Wilkins**

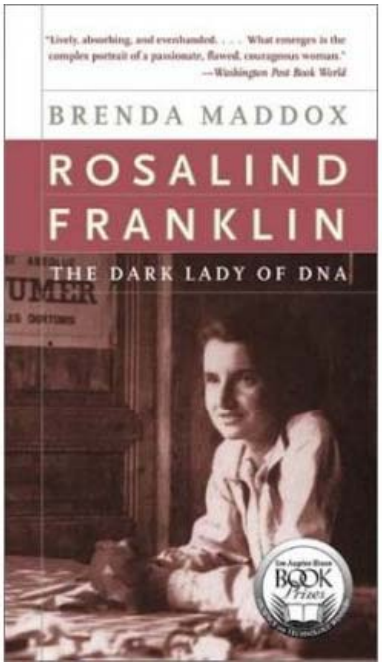
1/3 of the prize

United Kingdom and New Zealand

London University
London, United Kingdom

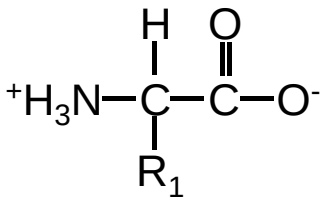




# Amino Acids

- **Amino acids**: the basic structural units of proteins. An amino acid is a compound that contains at least one amino group (-NH2) and at least one carboxyl group (-COOH)



Chang, 25.4

TABLE 25.1 The 20 Amino Acids Essential to Living Organisms*

| Name | Abbreviation | Structure |
|---|---|---|
| Alanine | Ala | |
| Arginine | Arg | |
| Asparagine | Asn | |
| Aspartic acid | Asp | |
| Cysteine | Cys | |
| Glutamic acid | Glu | |
| Glutamine | Gln | |
| Glycine | Gly | |
| Histidine | His | |
| Isoleucine | Ile | |

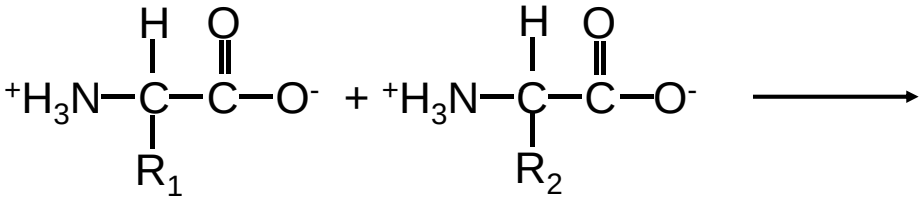
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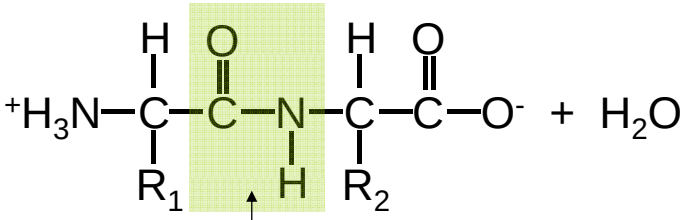
*The shaded portion is the R group of the amino acid.

9

# Proteins

- **Proteins**: A polymer of amino acids through peptide bonds.





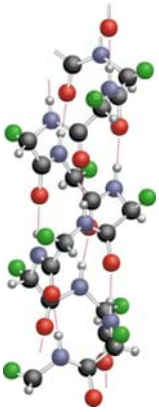
Chang, 25.4

**Condensation reaction**

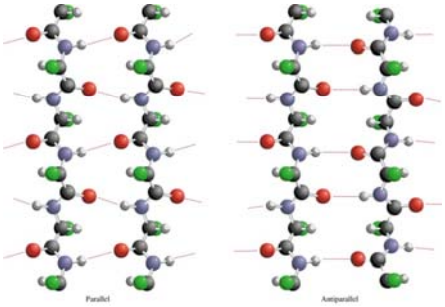
10

# Protein Structure

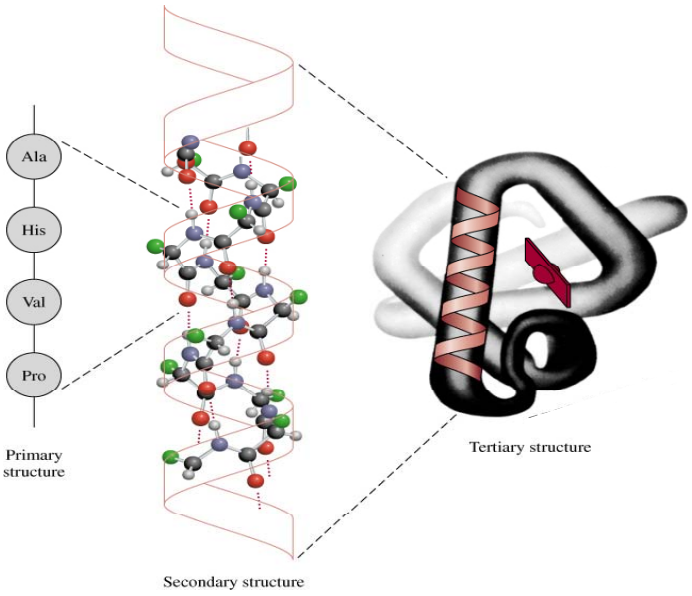
- α helix



- β sheet



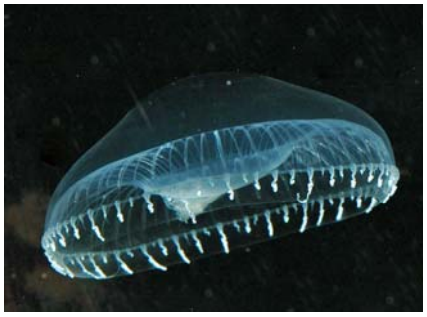
- Primary, secondary, and tertiary structures



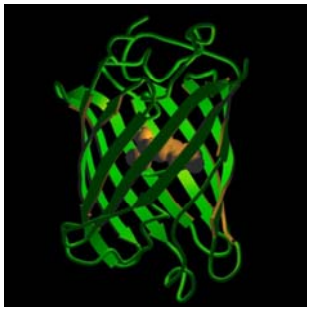

Chang, 25.4



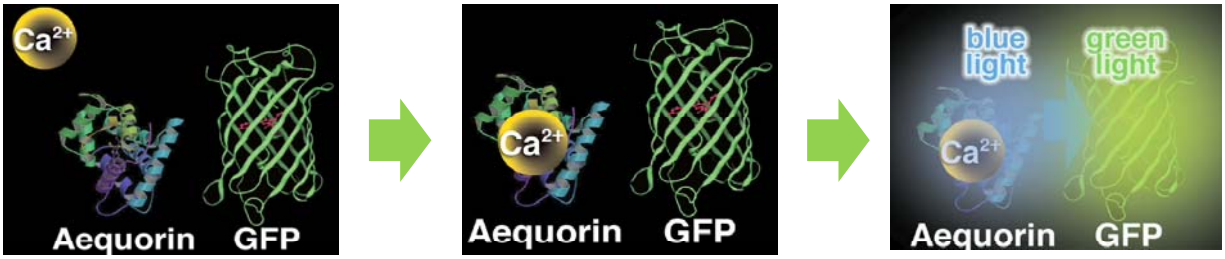
# What is Green Fluorescent Protein (GFP)?



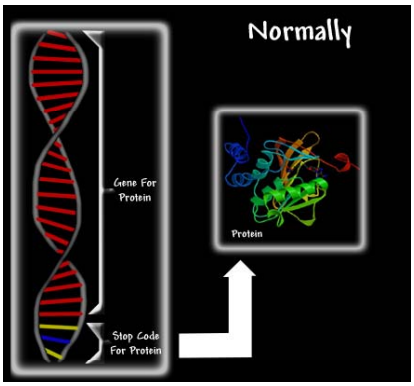
Aequorea victoria jellyfish

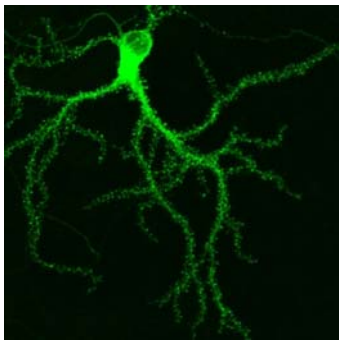


GFP

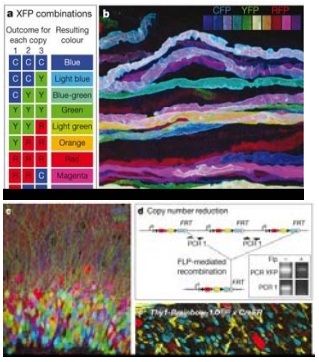


# Applications of GFP

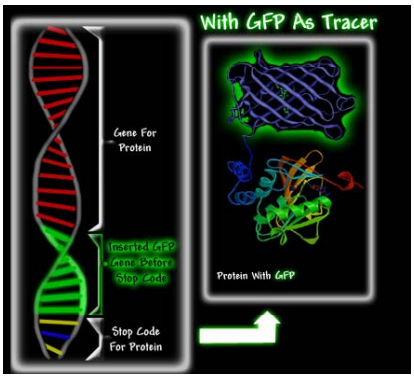


Neuron GFP

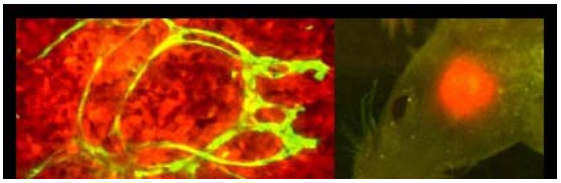



Livet et al., Nature **450**, 56 (2007)






GFP cloned 쥐와 보통 쥐



암세포들 (붉은색)과 혈관 (녹색)

http://www.conncoll.edu/ccacad/zimmer/GFP-ww/GFP-1.htm

KNU

## The Nobel Prize in Chemistry 2008

"for the discovery and development of the green fluorescent protein, GFP"

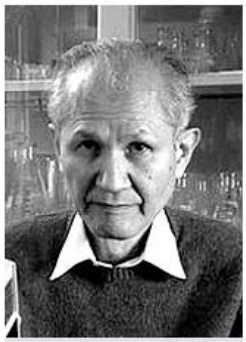


Photo: J. Henriksson/SCANPIX

**Osamu Shimomura**

1/3 of the prize

USA

Marine Biological Laboratory (MBL)
Woods Hole, MA, USA

b. 1928



Photo: J. Henriksson/SCANPIX

**Martin Chalfie**

1/3 of the prize

USA

Columbia University
New York, NY, USA

b. 1947



Photo: UCSD

**Roger Y. Tsien**

1/3 of the prize

USA

University of California
San Diego, CA, USA

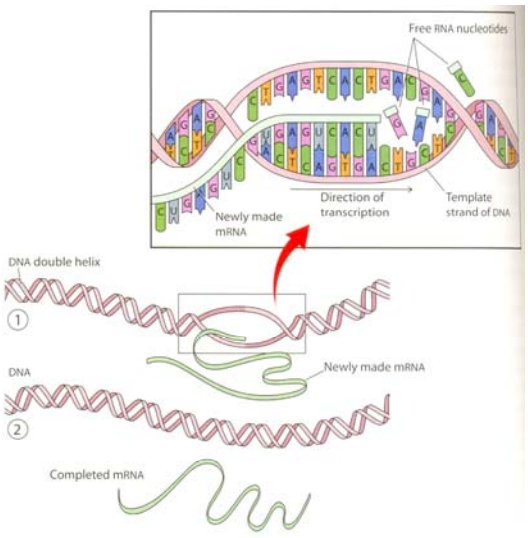
b. 1952

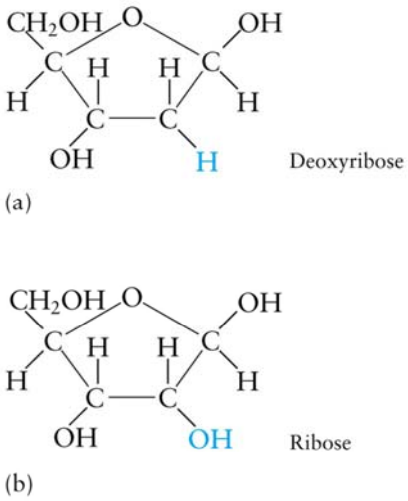


KNU

# Protein Synthesis: Transcription

- DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ protein
- Transcription takes place in the cell nucleus

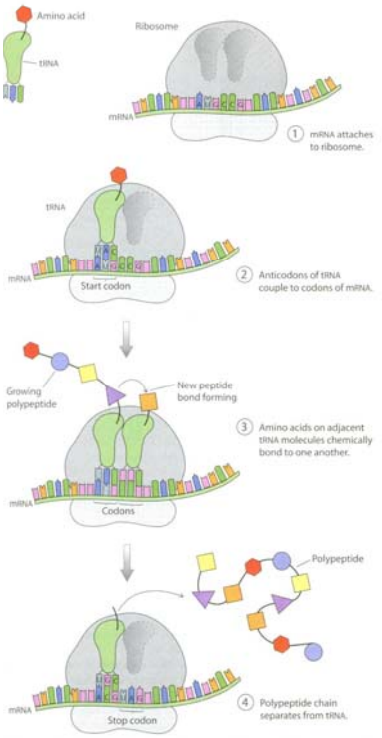


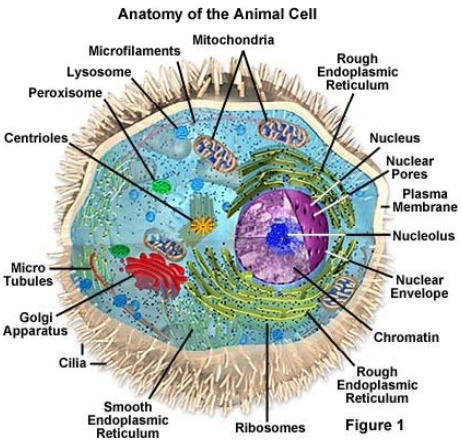



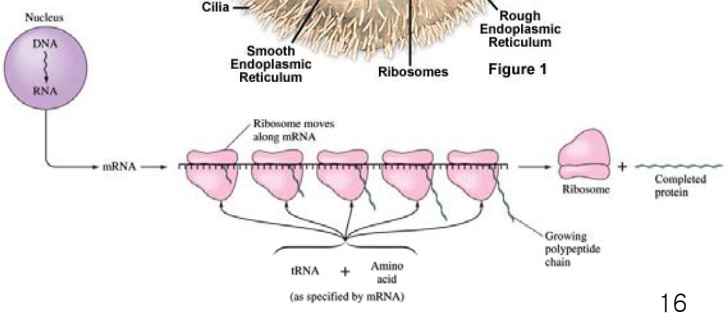



# Protein Synthesis: Translation

- Ribosomes are microscopic complexes of two-thirds rRNA and one-third proteins, and they are the site where proteins are built.



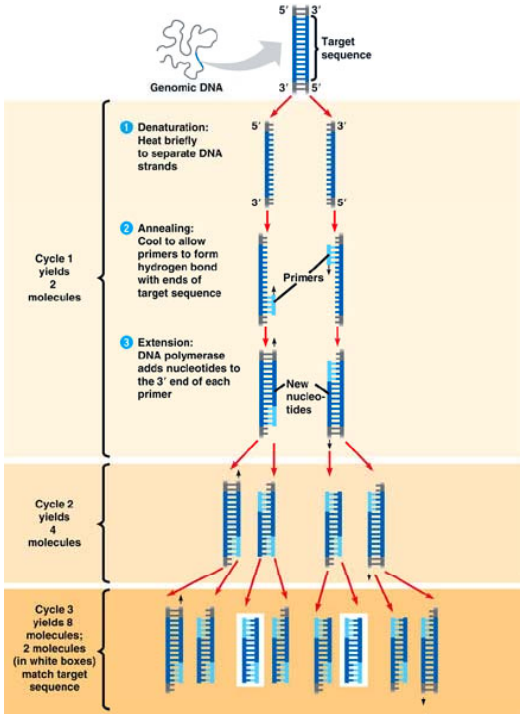



# Polymerase Chain Reaction (PCR)

- **PCR**
  - A technique to amplify a piece of DNA
  - DNA template (target), two primers, DNA polymerase (Taq polymerase), deoxinucleoside triphosphates.

- Procedure
  - Initialization: heating to 94-96°C
  - Denaturation: yielding ssDNAs
  - Annealing: lowering to 50-65°C and annealing primers to ssDNAs
  - Elongation: inducing DNA polymerases to synthesize a new DNA strand

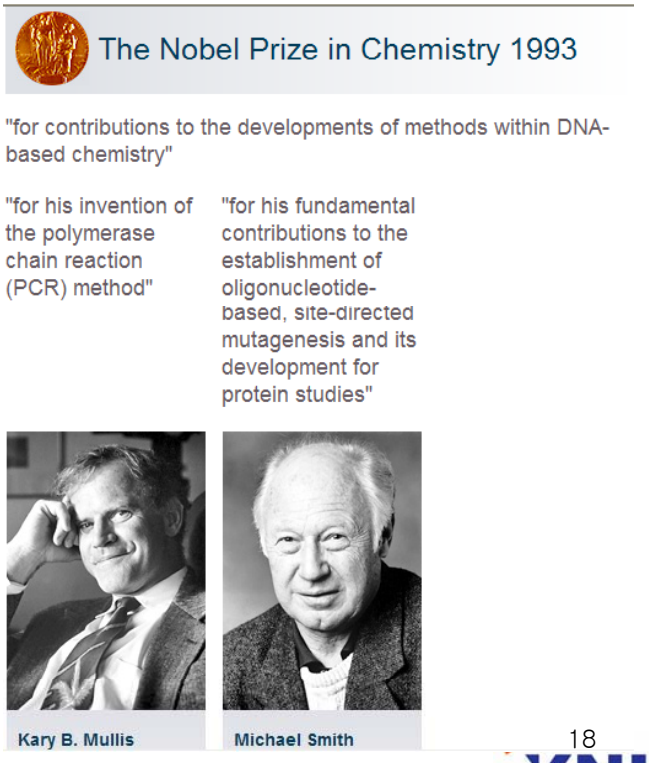
- Applications of PCR
  - Isolation of genomic DNA
  - DNA sequencing
  - Forensic analysis
  - Disease diagnosis



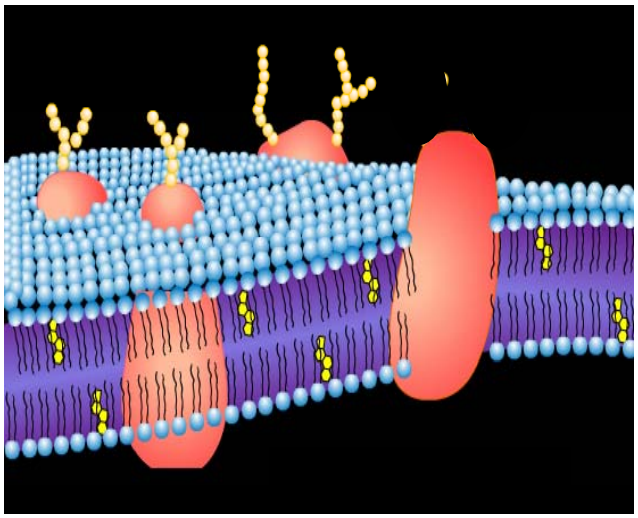



# Polymerase Chain Reaction (PCR)

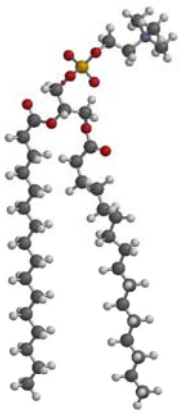


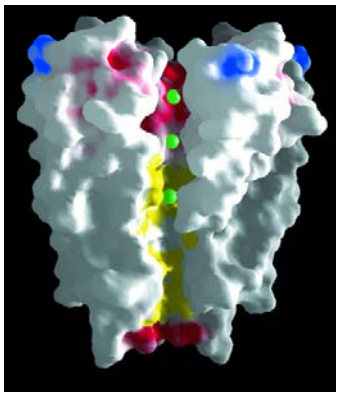



# What are ion channels?



Voet and Voet, Biochemistry (3rd. Ed.)





KcsA ion channel
Science **280,** 69 (1998)



Chemistry

## The Nobel Prize in Chemistry 2003

"for discoveries concerning channels in cell membranes"

"for the discovery of water channels"

"for structural and mechanistic studies of ion channels"



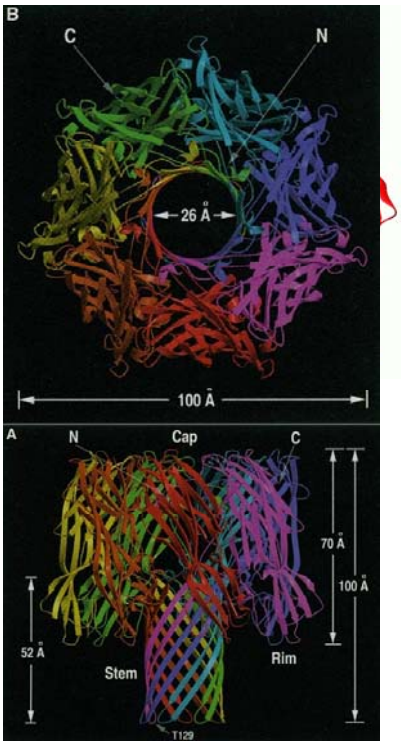
Peter Agre



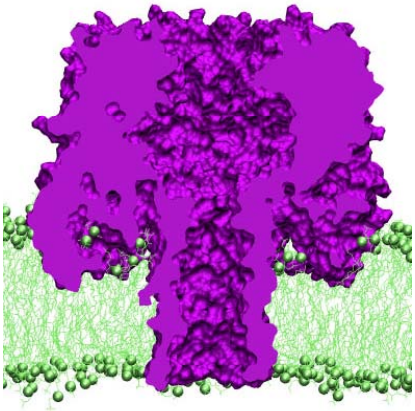
Roderick MacKinnon

# DNA translocation

- Alpha hemolysin: an ion channel

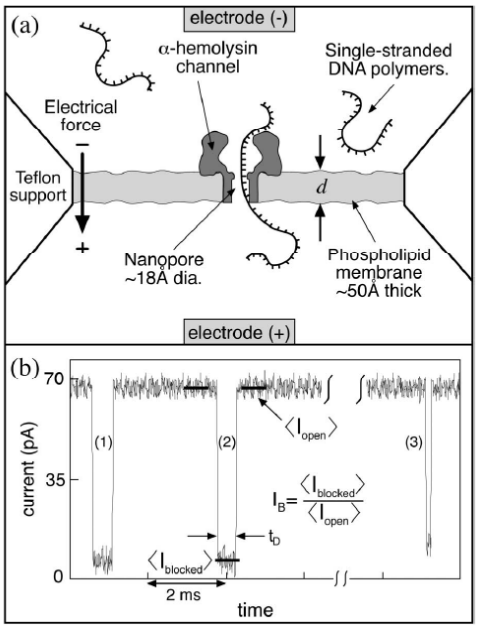



- Alpha hemolysin inside a lipid bilayer

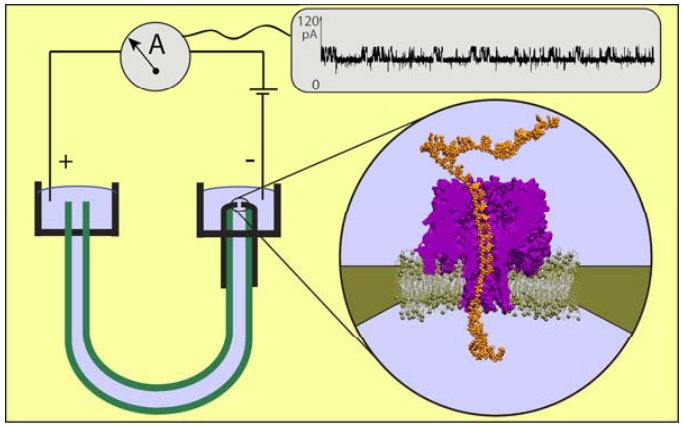




# DNA translocation

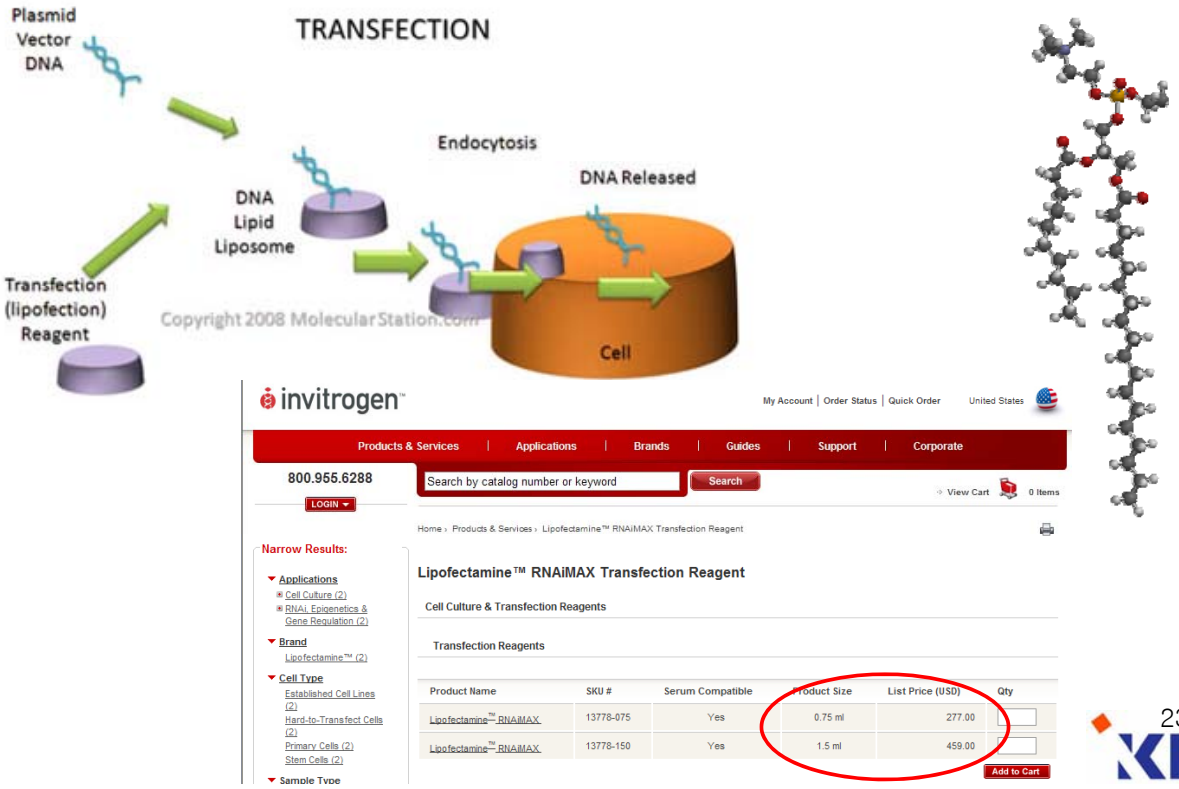



Miller et al., PRL **86**, 3435 (2001)




Schulten et al., PNAS **102**, 12377 (2005)

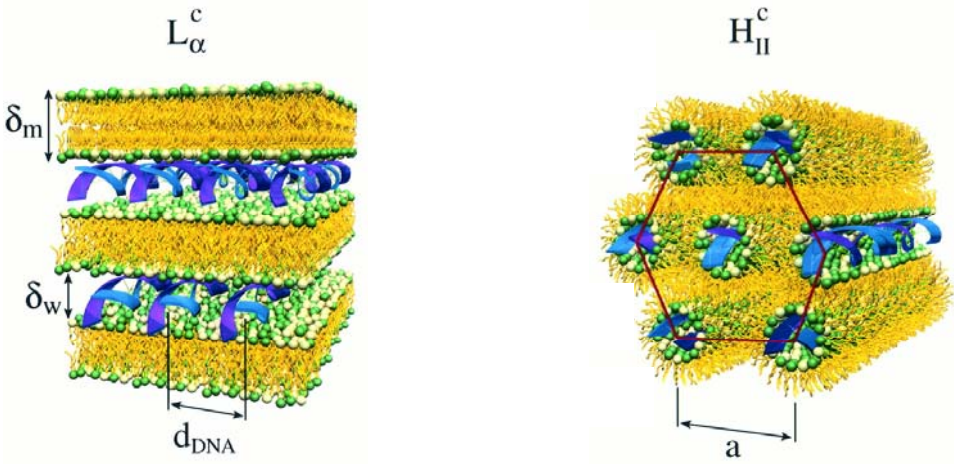
# DNA Transfection






# Lipid phases in lipid-DNA complex (=lipoplex)

- Lamellar($L_\alpha^c$) and hexagonal($H_{II}^c$) phases.




(Science, **281**, 78 (1998))

# Beer-Lambert Law

- Derivation of Beer-Lambert Law

$$dI = -\kappa[J]Idl \Rightarrow \frac{dI}{I} = -\kappa[J]dl$$

$$\int_{I_0}^{I} \frac{dI}{I} = -\kappa[J]\int_0^l dl \Rightarrow \ln\frac{I}{I_0} = -\kappa[J]l$$

$$\ln\frac{I}{I_0} = -\kappa[J]l \Rightarrow \log\frac{I}{I_0} = -\log e\kappa[J]l = -\varepsilon[J]l$$

- Absorbance

$$\boxed{A = \log\frac{I_0}{I} = \varepsilon[J]l}$$

Atkins, Physical Chemistry (8th ed.) 432-433

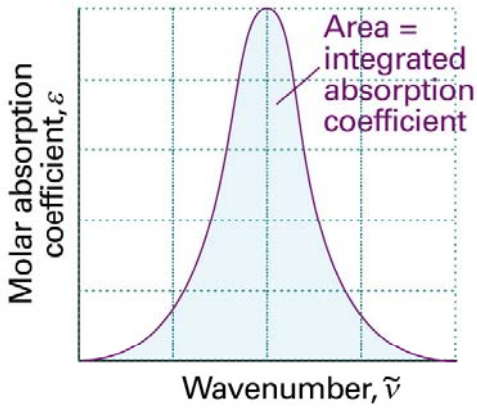



**Fig. 13.4** The integrated absorption coefficient of a transition is the area under a plot of the molar absorption coefficient against the wavenumber of the incident radiation.



# Colors and Complementary Colors

**The colors of the visible light**

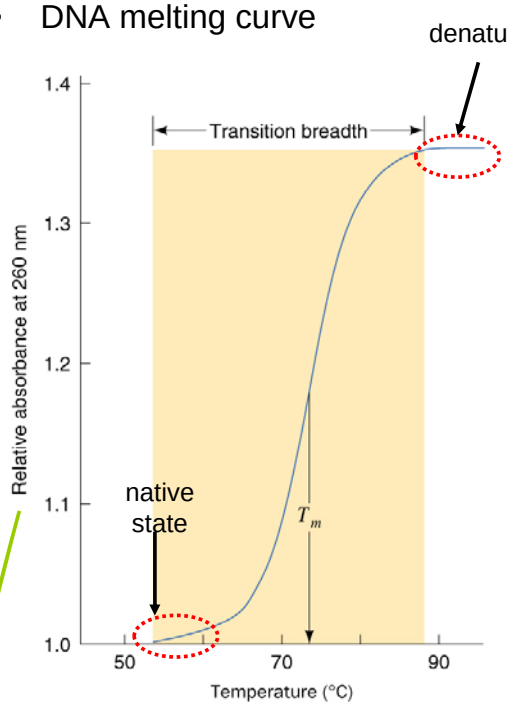
| color | wavelength interval | frequency interval |
|---|---|---|
| red | 700–635 nm | 430–480 THz |
| orange | 635–590 nm | 480–510 THz |
| yellow | 590–560 nm | 510–540 THz |
| green | 560–490 nm | 540–610 THz |
| blue | 490–450 nm | 610–670 THz |
| violet | 450–400 nm | 670–750 THz |

**Complementary colors**



# Hyperchromic effect of DNA

- DNA melting curve



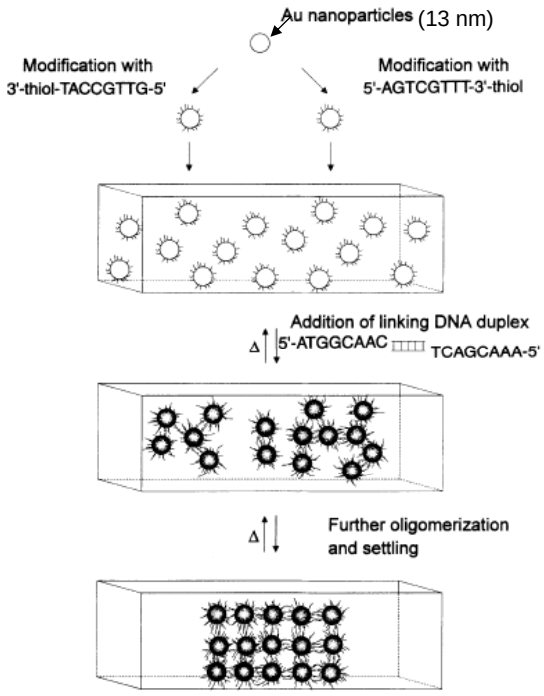

- Extinction coefficients of ss- and dsDNA at 260 nm
  1. For ssDNA: 0.027 $(\mu g/ml)^{-1}cm^{-1}$
  2. For dsDNA: 0.020 $(\mu g/ml)^{-1}cm^{-1}$

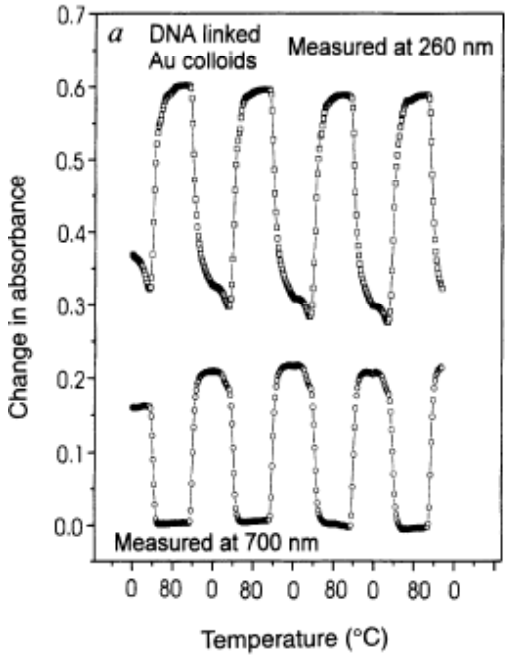
- Hyperchromic effect of DNA: when DNA denatures, its UV absorbance, which is almost entirely due to its aromatic bases, increases by ~40 % at all wavelengths.

Relative absorbance: the ratio of the absorbance at the indicated temperature to that at 25°C.



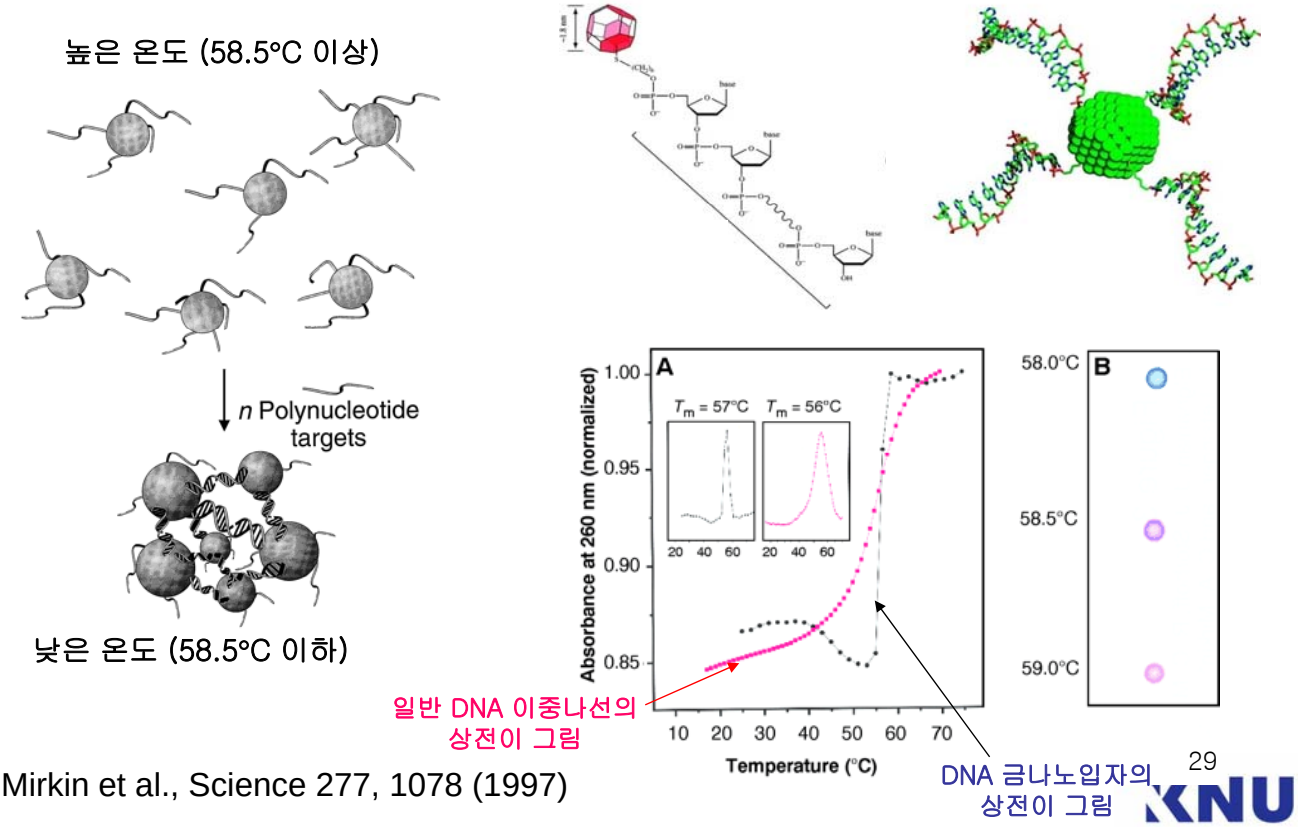
# Selective Colorimetric Detection of DNA




Mirkin et al., Nature **382**, 607 (1996)

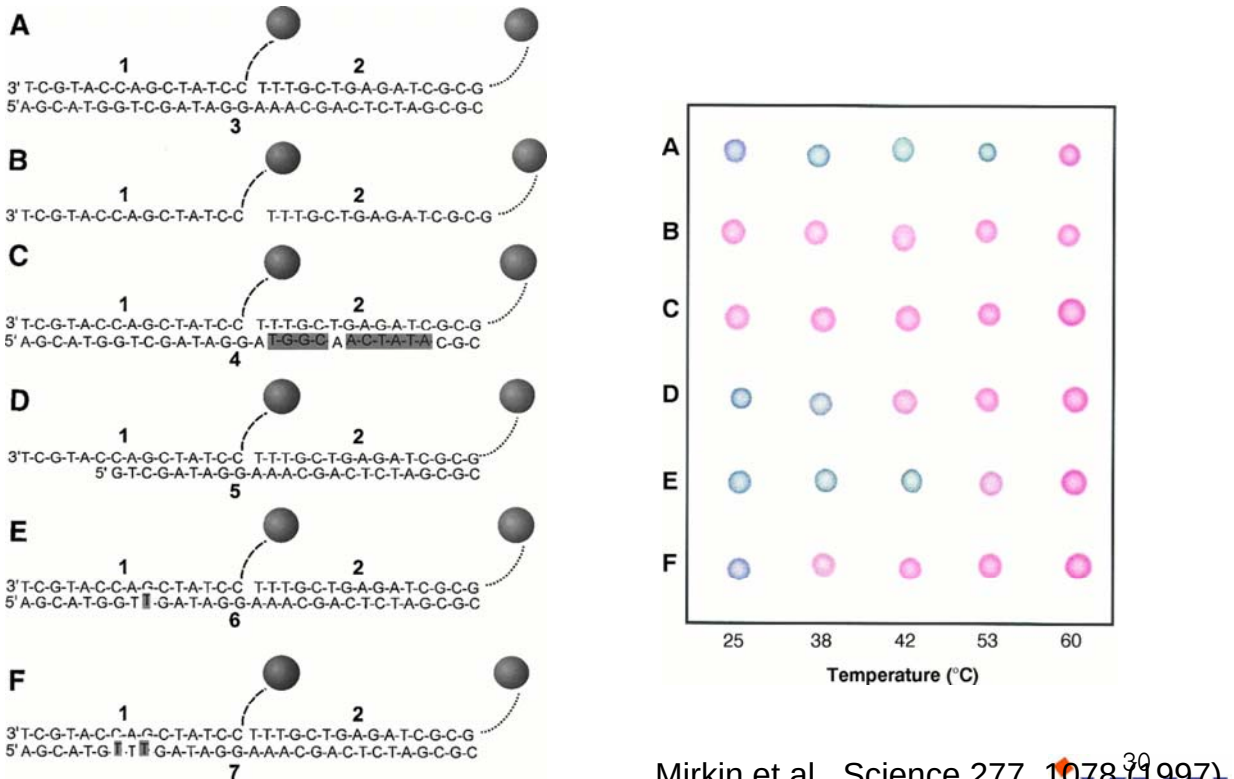
# Selective Colorimetric Detection of DNA




Mirkin et al., Science 277, 1078 (1997)

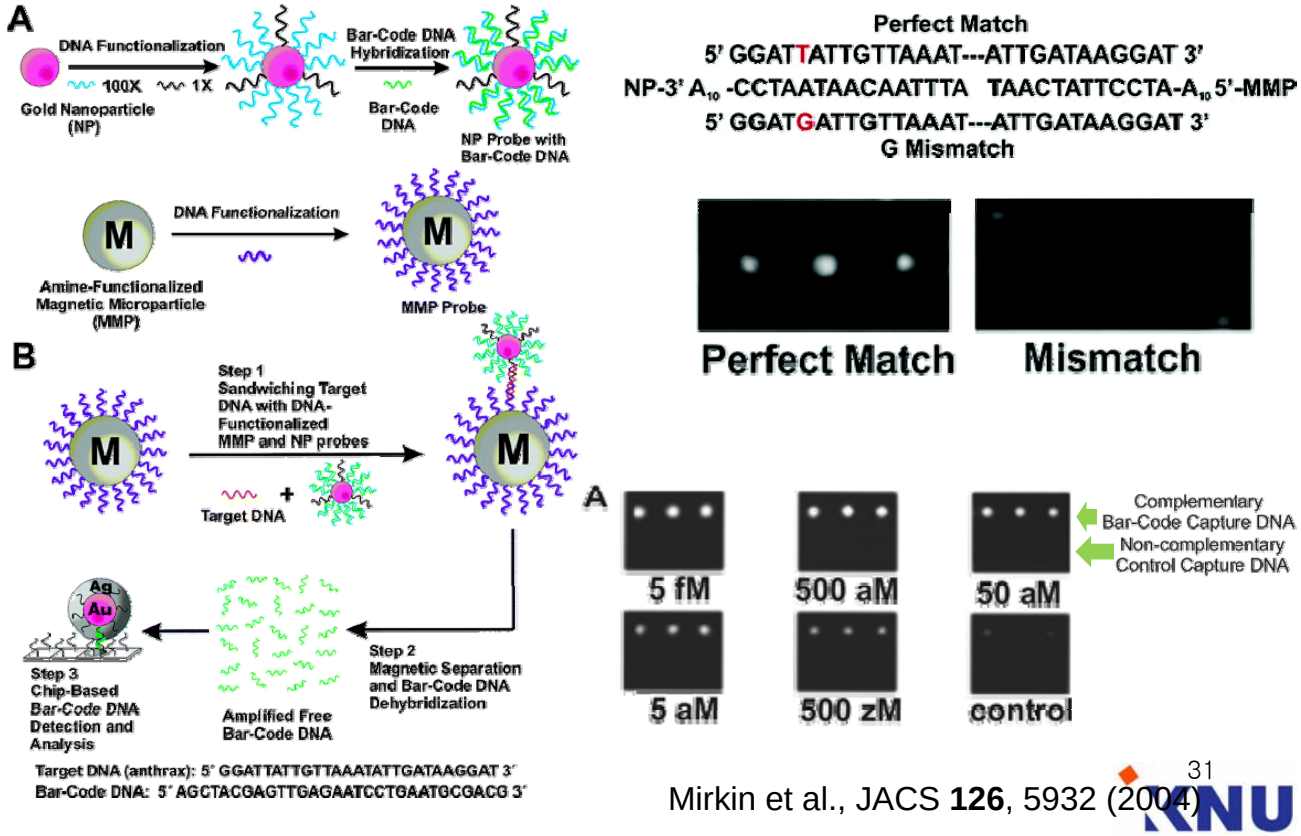


# Selective Colorimetric Detection of DNA




Mirkin et al., Science 277, 1078 (1997)

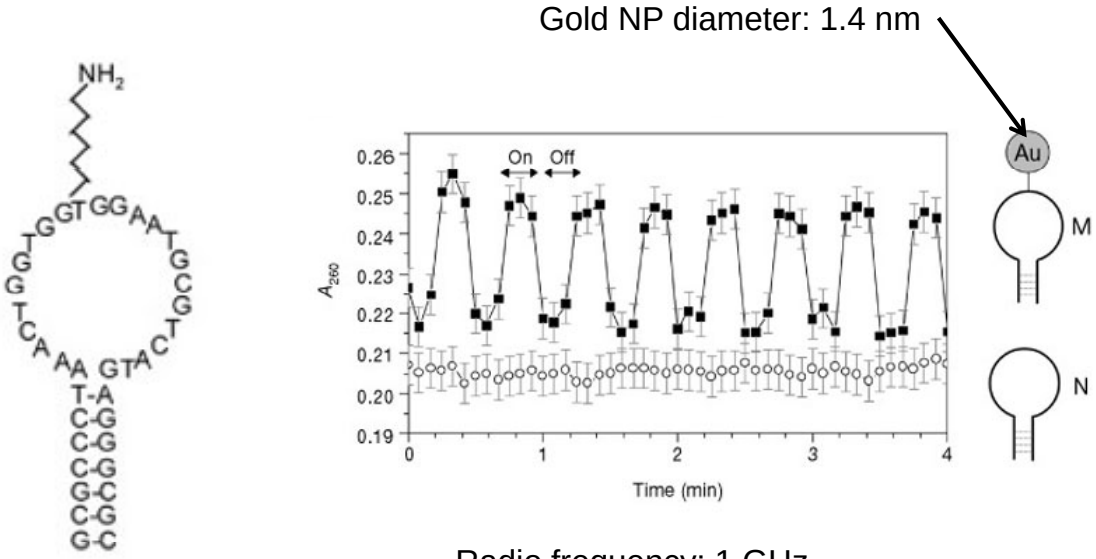
# Bio-Bar-Code-Based DNA Detection




Mirkin et al., JACS **126**, 5932 (2004)



# Remote Electronic Control of DNA Hybridization through Nanoscale Antenna

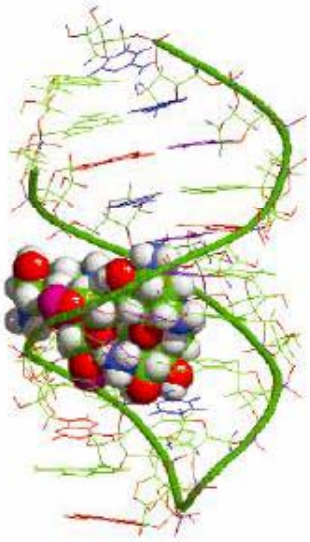


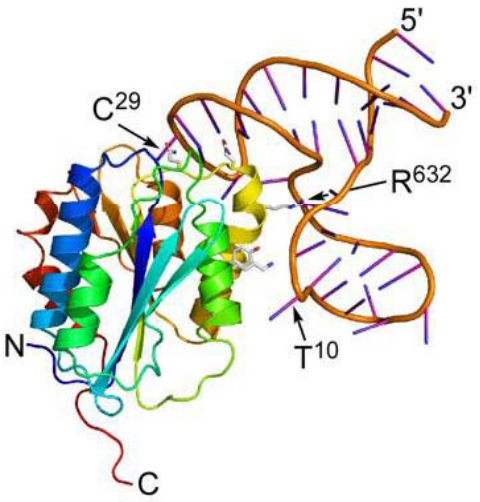

Radio frequency: 1 GHz

Hamad-Schifferli et al., Nature **415**, 152 (2002)

# Aptamer

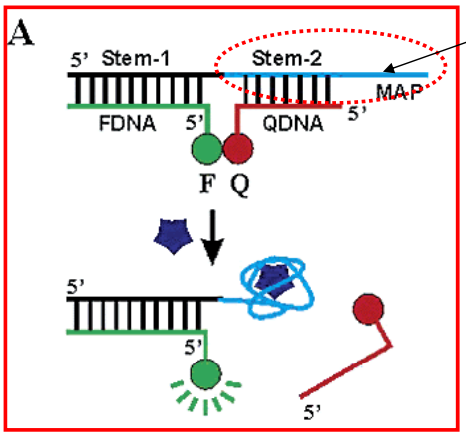
**Aptamer**: 핵산, 단백질, 금속이온과 같은 생체 내의 주요물질에 높은 결합력과 선택성을 가진 단일사슬 핵산 (single-stranded DNA)이나 단백질로서, 단백질로 이루어진 항체와는 달리 화학적, 물리적, 생물학적 안정성이 크다. (aptamer means “to fit” in Greek)



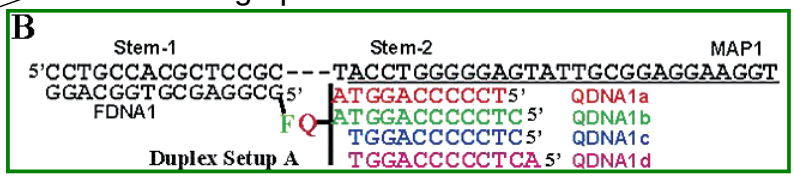


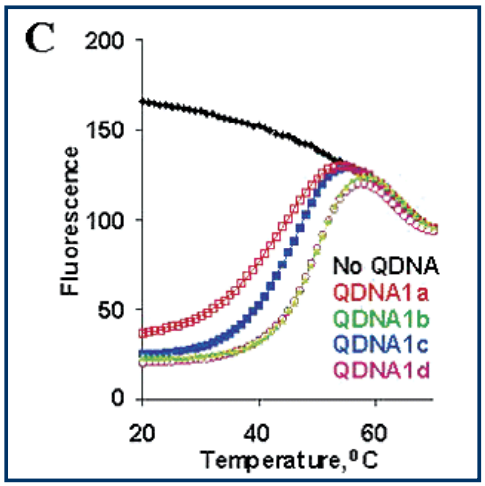


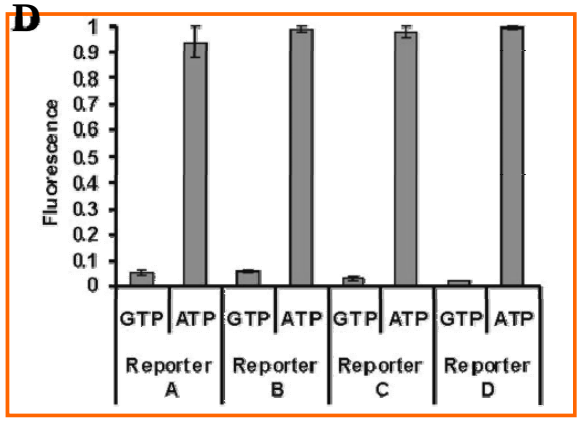

# Colorimetric Sensing of Adenosine Based on Aptamers




ATP binding aptamer

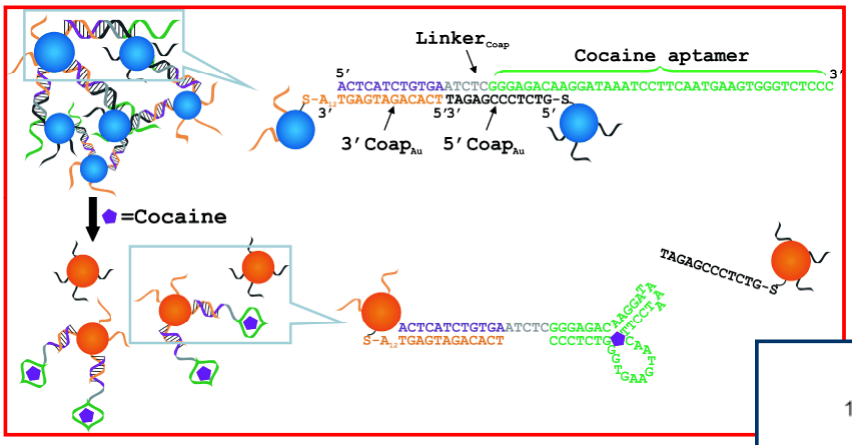


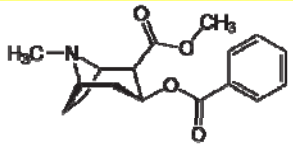




Nutiu and Li, JACS **125**, 4771 (2003)

# Colorimetric Sensing of Cocaine Based on Aptamers Attached to Gold Nanoparticles

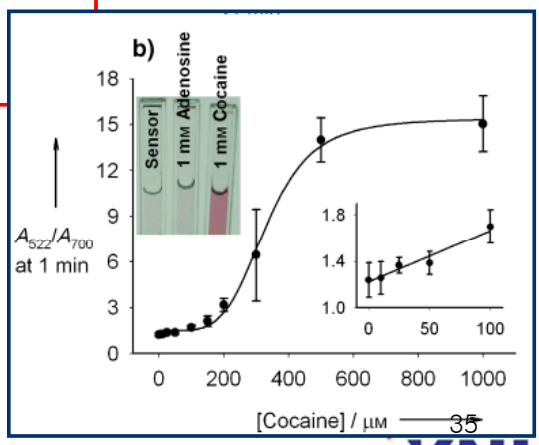



cocaine

DNA 금나노입자가 뭉쳐있을 때 → 700 nm 흡수, 파랑 또는 보라색

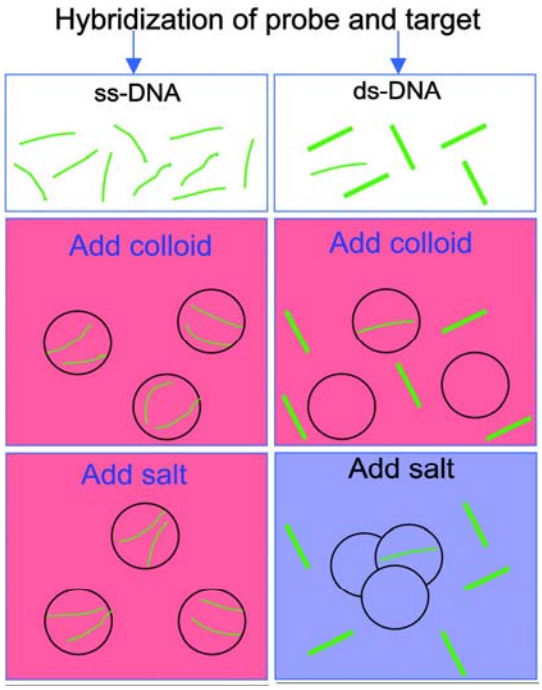
DNA 금나노입자가 떨어져 있을 때 → 522 nm 흡수, 붉은색

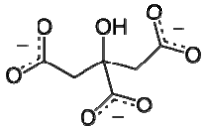



Liu and Lu, Angew. Chem. Int. Ed. **45**, 90 (2006)

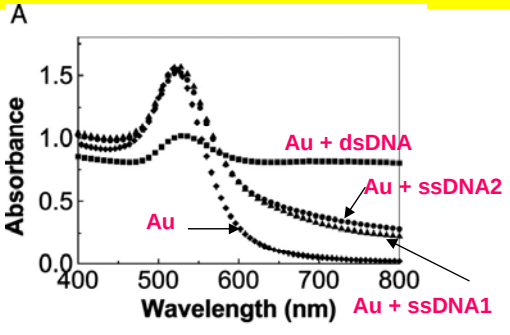


# Colorimetric detection of DNA sequences based on electrostatic interactions



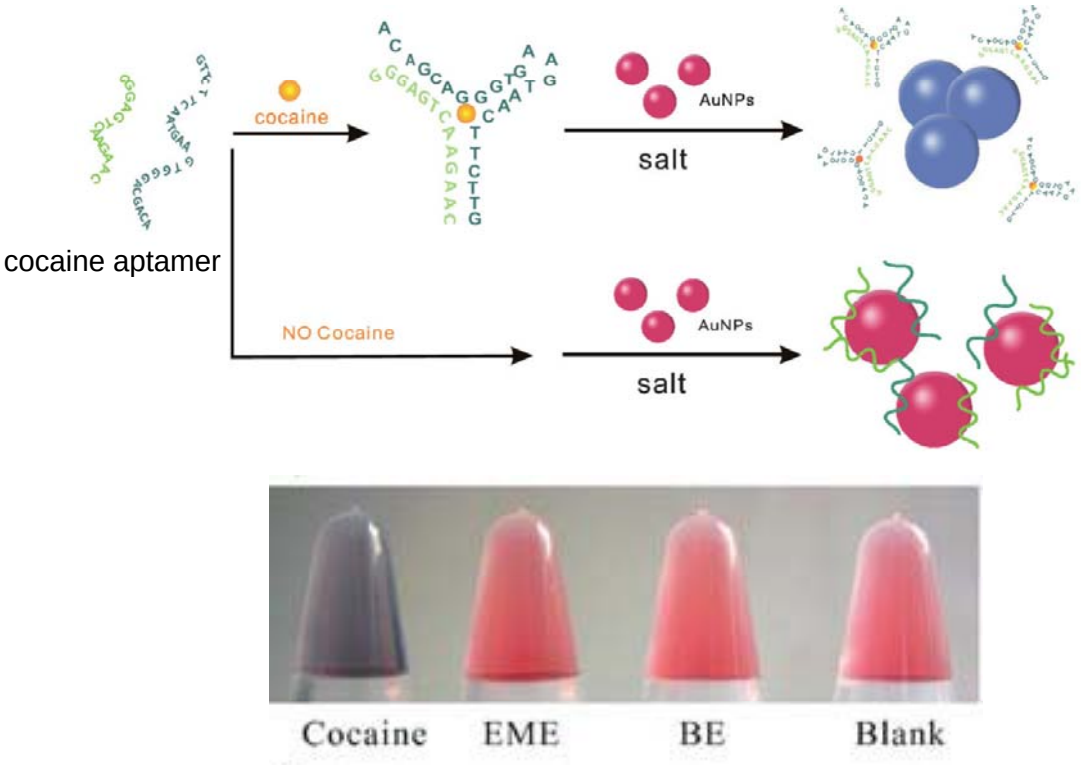




citrate




Li & Rothberg, PNAS **101**, 14036 (2004)

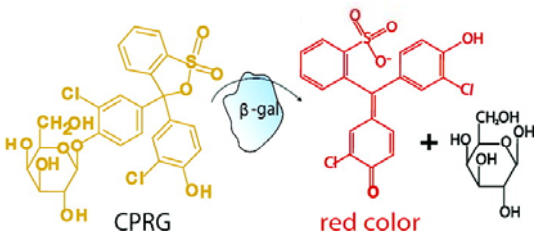
# Colorimetric Sensing of Cocaine Based on Aptamers and Gold Nanoparticles




Zhang et al., Small **4**, 1196 (2008)



# Colorimetric Sensing of Bacteria Using Enzyme-Nanoparticle Biosensor




β-galactosidase (β-gal) :효소
CPGR = β-D-galactopyranoside: 기질

